

Assignment Fourier Series

1. Find a Fourier series to represent $x - x^2$ from $x = -\pi$ to $x = \pi$.

2. An alternating current after passing through a rectifier has the form

$$I = \begin{cases} I_0 \sin x & \text{for } 0 \leq x \leq \pi \\ 0 & \text{for } \pi \leq x \leq 2\pi \end{cases}$$

Where I_0 the maximum current and the period is 2π . Express i as a Fourier series.

3. Let f be the 2π -periodic function which satisfies $f(x) = \begin{cases} \pi + x & \text{for } -\pi < x < 0 \\ \pi - x & \text{for } 0 < x < \pi \end{cases}$
Find the Fourier series of f and hence deduce the value of $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$.

4. Find the Fourier series of the periodic function $f(x) = \begin{cases} -k, & -\pi \leq x \leq 0 \\ k, & 0 \leq x \leq \pi \end{cases}$ and $f(x + 2\pi) = f(x)$. Sketch the graph of $f(x)$ and deduce that $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2n-1} = \frac{\pi}{4}$.

5. Obtain a half range sine series for $f(x) = \begin{cases} kx, & 0 \leq x \leq \frac{1}{2} \\ k(l-x), & \frac{1}{2} \leq x \leq 1 \end{cases}$.

6. Obtain a half range cosine series for $f(x) = x \sin x$ in the interval $(0, \pi)$.

7. Find the complex form of the Fourier series of $f(x) = e^{-x}$ in $-1 \leq x \leq 1$.

8. Find the complex form of the Fourier series of $f(x) = \cos ax$ in $-l \leq x \leq l$.

9. Compute the constant term and the first harmonic of the Fourier series of $f(x)$ given in the following table:

x	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$	2π
y	1.0	1.4	1.9	1.7	1.5	1.2	1.0

10. The following data gives the variation of periodic current over a period

t	0	$T/6$	$T/3$	$T/2$	$2T/3$	$5T/6$	T
A	9.0	18.2	24.4	27.8	27.5	22.0	9.0

Find numerically the direct current part of the variable current and obtain the amplitude of the first harmonic.

